

$Z \rightarrow \tau_h \tau_h$ cross section with CMS Open Data

Methods, choices and findings

BND School 2026 – $Z \rightarrow \tau\tau$ subgroup

14 September 2026 • branch z-tautau

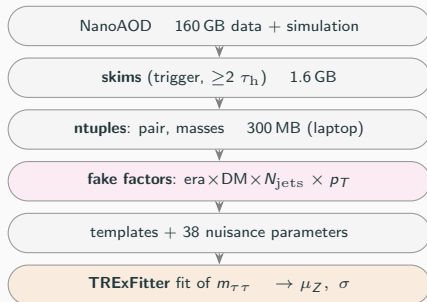
The measurement at a glance

Goal: $\sigma(pp \rightarrow Z/\gamma^* \rightarrow \tau\tau)$ with both τ decaying hadronically, in the same TRExFitter framework as ee and $\mu\mu$, ready for a combined fit.

Data: CMS 2016 Tau dataset, Run2016G+H (UL NanoAODv9), 156 M events, $L = 16.39 \text{ fb}^{-1}$ (normtag, same in all channels)

Strategy

- two τ_h ($p_T > 40 \text{ GeV}$), OS, DeepTau Medium: 47 586 events
- 81 % jet $\rightarrow\tau_h$ fakes $\Rightarrow$ fake factor from data, no QCD MC
- $Z \rightarrow \tau\tau$ + small backgrounds from UL16 simulation, TauPOG corrections
- MET folded into the visible mass: likelihood mass $m_{\tau\tau}$
- binned profile-likelihood fit of $m_{\tau\tau}$, TRExFitter v1.8.0 (shared binary)



$$\sigma(Z/\gamma^* \rightarrow \tau\tau, 60\text{--}120) = 2255 \pm 36^{+371}_{-315} \pm 83 \text{ pb}$$
$$\text{NNLO: } 1945 \text{ pb} \quad \mu_Z = 1.16^{+0.19}_{-0.16}$$

Selection: choices and why

trigger	DoubleMedium(Combined)IsoPFTau35.Trk1_eta2p1_Reg:
	G has only <i>Iso</i> , H only <i>CombinedIso</i> ; both legs matched to HLT τ objects
τ_h	$p_T > 40$ (plateau side of the 35 GeV leg), $ \eta < 2.1$, DM 0/1/10/11 (calibrated modes)
DeepTau 2017v2p1	VSe VVLoose, VS μ VLoose; VSjet Medium = tight, VVVLoose $\wedge\neg$ Medium = loose
pair	most isolated, $\Delta R > 0.5$; τ_1 = leading p_T (FF leg)
veto	no extra e/μ : orthogonal to $e\tau_h$, $\mu\tau_h$, ee , $\mu\mu$
simulation	drop events whose leading τ is a jet (covered by the FF)

cutflow	Run2016G	Run2016H
NanoAOD	79.6 M	76.8 M
skim	2.81 M	2.22 M
pair + MET filters	2.03 M	1.62 M
trigger match	1.99 M	1.59 M
lepton veto	1.99 M	1.59 M
SR (OS, Medium)	24 106	23 480

Expected SR composition

fakes 38 746 $Z \rightarrow \tau\tau$ 7 878

$t\bar{t}$ 570, W+jets 514, $Z \rightarrow ee$ 208, VV 122

Signal acceptance is tiny: $A = 0.23\%$

($B_{hh} \approx 42\% \times$ both visible τ above 40 GeV)

MET correction of the visible di- τ mass

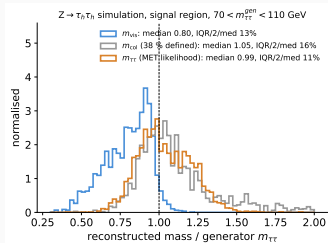
≥ 4 neutrinos escape: m_{vis} peaks at $0.8 m_Z$.

- **collinear approx.:** solve MET

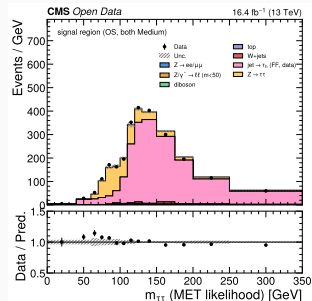
$$= \sum_i (1/x_i - 1) \vec{p}_{T,i}^{\text{vis}}$$
 singular for back-to-back τ : **defined in 38%**
- **likelihood mass** $m_{\tau\tau}$ (FastMTT/SVfit-lite idea): scan (x_1, x_2) on a grid,
 weight = $\exp(-\frac{1}{2} r^T V^{-1} r)$ (MET covariance)
 $\times$ flat 2-body phase space $\times 1/m^2$
 $\rightarrow$ posterior median, always defined

	scale	resolution
m_{vis}	0.80	13 %
$m_{\tau\tau}$ (used)	0.99	11 %

$Z \rightarrow \tau\tau$ /fakes in the best window: 2.6 ($m_{\tau\tau}$) vs 2.3 (m_{vis})



Finding: correct scale and better resolution;
 separation power vs fakes about equal. With
 $p_T > 40$ on both legs the selected τ are
 already hard.



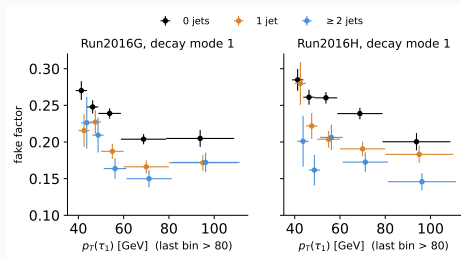
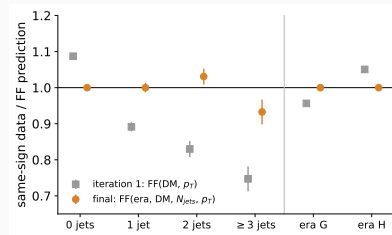
Signal region, prefit: Z peak at m_Z , fakes
 at 130–300 GeV give the normalisation
 sideband.

Jet $\rightarrow \tau_h$ fakes: the classic fake factor

$$FF = \frac{N(SS, \tau_1 \text{ Medium}, \tau_2 \text{ Medium})}{N(SS, \tau_1 \text{ VVVL} \wedge \neg M, \tau_2 \text{ Medium})}$$

$$N_{\text{fake}}^{\text{SR}} = \sum_{OS, \tau_1 \neg M, \tau_2 M} C_{OS/SS} \cdot FF$$

- same-sign = QCD enriched (no Z); τ_2 Medium in both $\Rightarrow$ isolation correlation absorbed
- binning chosen by **closure** in same-sign data:
 - N_{jets} : off by +9 % . . . - 25 % (quark/gluon mix)
 - era: -4 % (G) / +5 % (H): different HLT τ isolation
- final: era $\times$ DM $\times N_{\text{jets}}$ (0,1, ≥ 2) $\times p_T = 120$ bins, $\sim 10\%$ stat. each
- $C_{OS/SS} = 1.07\text{--}1.13$ from the τ_2 anti-isolated sideband, +3 % extrapolation syst.



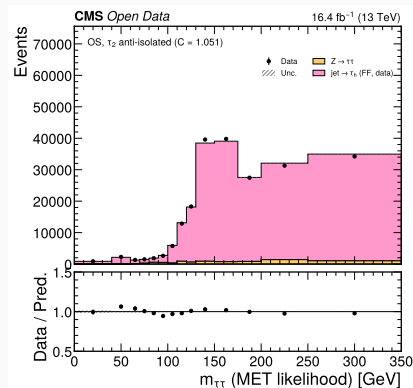
all 120 bins: output/plots/step3_fakefactors.png

The MC-subtraction question

Genuine τ leak into the fake-factor regions (simulated, genuine τ_1):

region	data	genuine τ_1	fraction
SS, τ_1 Medium (FF num.)	26 761	232	0.9 %
OS, τ_1 fails (application)	187 973	2 386	1.3 %
OS, τ_2 anti-iso (C num.)	219 737	8 321	3.8 %

- **Nominal (as specified): no subtraction.** Signal is double counted in the application region and C is biased up.
- Alternative “mcsb” computed everywhere: prefit fakes $-4.5\% = 22\%$ of the signal; $C: 1.078 \rightarrow 1.051$
- But post-fit: $\mu_Z = 1.160$ vs 1.208 — the high- $m_{\tau\tau}$ sideband re-normalises the fakes (Fake0SSS pulled -0.86σ towards mcsb)
- **Recommendation:** make “mcsb” nominal next



OS, τ_2 anti-isolated: after C and subtraction the SS $\rightarrow$ OS shape closes within $\pm 5\%$.

TauPOG UL2016 postVFP (public GitHub; cvmfs/GitLab not reachable)

	DM0	DM1	DM10	DM11
ID SF (Medium)	0.92(14)	0.88(5)	0.87(9)	0.90(16)
energy scale	0.993(9)	0.991(7)	1.001(7)	0.997(16)
trigger SF @50	0.94(5)	0.97(4)	0.94(5)	1.07(8)

+ $e/\mu \rightarrow \tau_h$ SFs, pileup (Open Data lumi table), L1 prefiring

38 nuisance parameters + 14 γ , names shared across channels:
Lumi, Pileup, L1Prefiring, QCDScale, PDF, PS_ISR/FSR,
SigModel (correlated); TauID/Trigger/ES_DM*,
MET_Unclustered, Fake*_tautau, XS_*

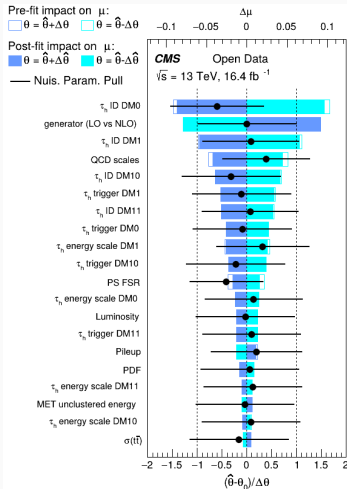
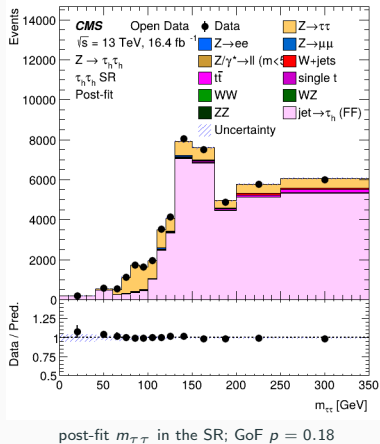
Choices that mattered

- theory variations *renormalised to the fiducial yield*: only C in the fit, A separate (3.7 %)
- LO vs NLO generator: normalisation only ($\Delta C = 7.3\%$)
- W+jets: few events, weights up to $\sim 200 \Rightarrow$ shape from $\tau_2 \geq \text{VVVLoose}$, 50 % norm. NP

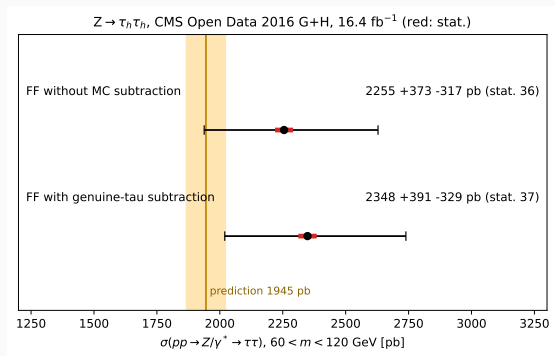
Bugs found and fixed

- scale envelope & PDF Hessian taken *per event*:
 $\pm 15\% / \pm 4\% \rightarrow$ per histogram: $\pm 4\% / \pm 1\%$
(σ_μ : 26% $\rightarrow$ 18%)
- forked worker pools deadlock after uproot; uproot ≥ 5.6 writes dicts as RNTuple

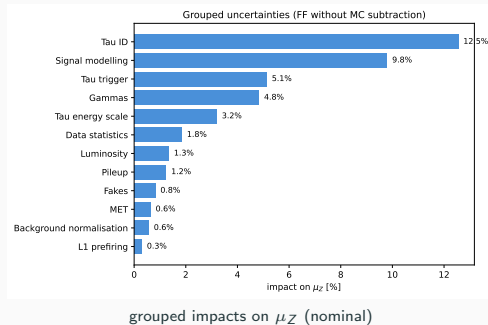
The fit



- one region, 14 bins of $m_{\tau\tau}$ (0–350 GeV)
- μ_Z on $Z \rightarrow \tau\tau$; fakes from data with FF nuisances
- bins above 130 GeV (95 % fakes) fix the fake normalisation: post-fit fakes –2.8 %
- pulls within 1σ ; FakeClosure constrained to 0.32
- uncertainty on μ_Z :
 expected $+0.169$
 -0.140 , observed $+0.192$
 -0.163
- stat-only: ± 0.018



$\sigma(Z/\gamma^* \rightarrow \tau\tau, 60 < m < 120)$	$2255 \pm 36^{+371}_{-315} \pm 83_A \text{ pb}$
prediction (FEWZ NNLO)	1945 pb
$\sigma_{\text{fid}}(\tau_h \tau_h, p_T^{\text{vis}} > 40, \eta < 2.1)$	$5.22 \pm 0.08 \pm 0.80 \text{ pb}$
prediction	4.50 pb
$\mu_Z \text{ (nominal / mcsb)}$	$1.160^{+0.192}_{-0.163} / 1.208^{+0.201}_{-0.169}$



Findings

1. Consistent with NNLO within 1σ ; data statistics 1.8 %, total $^{+17}_{-14}$ %.
2. **The limit is external:** TauPOG τ_h ID SFs (12.5 %) — $Z \rightarrow \tau\tau$ is how those SFs are measured.
3. Background is $5\times$ the signal; only a *shape* fit works (counting: 1815 pb, driven by a 3 % fake normalisation).
4. The FF needs N_{jets} and era binning to close; the MC-subtraction choice moves prefit fakes by 22 % of the signal but μ_Z by only 4 %.
5. The MET-likelihood mass fixes the scale and improves the resolution, but does not add discrimination at $p_T > 40$.

For the combination

- workspace fit/results/ztautau, POI mu_Z: plugs into fitting/combination_skeleton.config
- with $ee/\mu\mu$ fixing μ_Z , $\tau\tau$ constrains the τ_h ID NPs (lepton universality)

Next steps

- MC-subtracted FF as nominal; closure corrections in $\eta(\tau_1)$, $p_T(\tau_2)$
- NLO-only signal-model prescription (LO vs NLO: 8.5 %)
- more DY statistics (jet-binned), HT-binned W+jets
- add $e\tau_h/\mu\tau_h$: τ_h ID in situ

Code, docs, handoff: z-tautau/ (README, CLAUDE.md, docs/00-08)